BIG STACK PEOPLE | HRMS

Phase Two
Workforce & Payroll

Scope definition · implementation priorities · two-day delivery plan · acceptance criteria

OBJECTIVE	DELIVERY PRINCIPLE
Strengthen day-to-day workforce operations with shared holiday calendars, employee notifications, effective-dated work schedules, and transparent compensation records.	Keep permissions server-enforced, preserve history, validate every write, and make employee self-service useful on desktop and mobile.

Four workstreams

01 Calendar	02 Notifications	03 Shifts	04 Compensation
Holiday scopes, yearly/month views, attendance calendar rules.	Personal inbox, read state, preferences, scoped announcements.	Shift catalogue, effective assignments, attendance metrics and reports.	Salary components, effective-dated revisions, deductions and payroll history.

Repository baseline and two-day assumption

The current repository already contains Phase Two backend routes, React workspaces, and workflow tests for these four areas. This two-day timebox is therefore planned as integration, gap closure, regression verification, and handover—not as a promise to build every capability from scratch. Any missing requirement discovered during Day 1 review must be raised immediately and prioritized against the deadline.

Baseline evidence: backend/app/phase_two.py, frontend/src/PhaseTwo.jsx, and backend/tests/test_auth.py. Production readiness still depends on deployment configuration, data migration review, and stakeholder acceptance.

01 Holiday & calendar management

Maintain one authoritative holiday calendar that employees can browse and that attendance and leave calculations can consult consistently.

Capability	Expected behavior
Company-wide	A holiday applies across the organization.
Regional	A department-scoped holiday is visible only to matching employees.
Optional	Visible as an optional day; excluded from working-day totals only according to configured policy.
Calendar views	Employee-facing month and year views show dates, names, and scope.
Attendance summary	Non-optional applicable holidays are surfaced in monthly summaries and excluded from expected working-day counts.

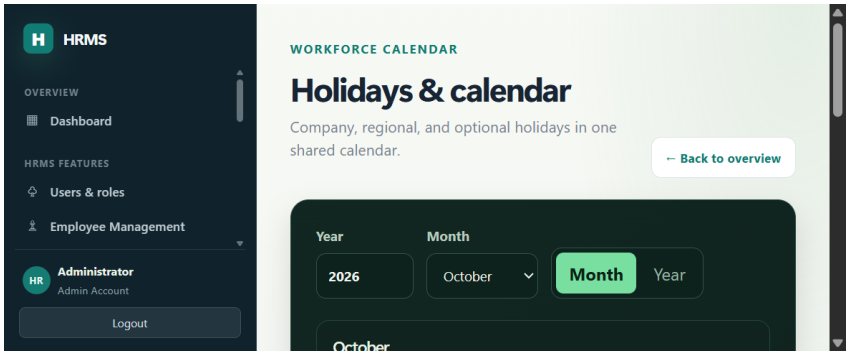
Primary workflow

Admin creates a dated holiday and selects company, regional, or optional scope. The system validates the date, scope, and department, prevents conflicting duplicate entries, and records the change. Employees see only holidays applicable to their organization or region. Attendance summaries use the same scope rules as the calendar.

Acceptance checks

- Company holidays are visible to all active employees.
- Regional holidays are restricted to employees in the selected department.
- Optional holidays are visibly marked and do not silently reduce required workdays.
- Month and year navigation show the correct dates, including leap years.
- Monthly attendance totals exclude applicable non-optional holidays and report holiday counts.
- Only authorized roles can create or remove calendar records.

Policy note: the organization should confirm whether optional holidays reduce attendance expectations when individually selected; the default plan does not assume they do.



Application output: live Holiday & Calendar screen (October 2026; no holidays configured in this demo database).

02 In-app notification center

Provide a persistent, user-specific inbox for operational events alongside existing email delivery. In-app notification delivery and email delivery are tracked as separate channels.

Event	Recipient / intent
Leave status	Notify the employee when a request is approved or rejected.
Payroll processing	Notify relevant employees when a pay run is generated or credited.
Attendance alert	Surface configured attendance exceptions to the affected employee or authorized manager.
Admin announcement	Deliver organization-wide or selected-department messages.

Inbox and controls

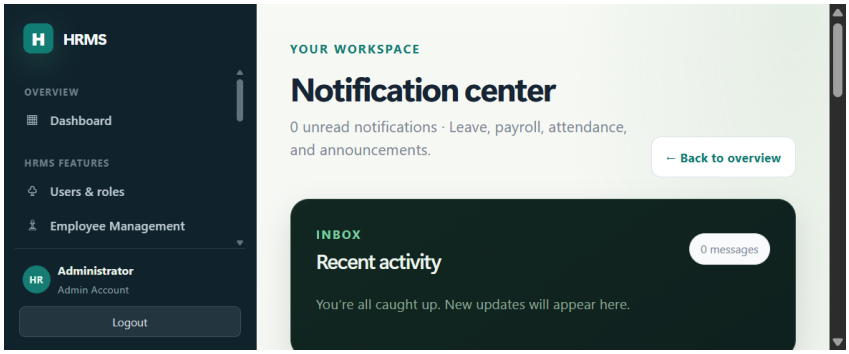
Each notification belongs to one user and includes an event type, title, message, creation time, and read state. The inbox shows unread count, supports marking an item read, and exposes per-event preferences. Turning off an event preference suppresses future in-app notices for that event; it does not alter historical records or silently change email settings.

Broadcast safeguards

Admin selects one or more departments, or sends to the whole organization. The server validates department IDs, removes duplicate recipients, and creates each notice only for an eligible active user. Broadcast actions are auditable and must not reveal the recipient list to other employees.

Acceptance checks

- Employees can see and update only their own inbox and preferences.
- Read/unread state and unread count remain consistent after refresh.
- Disabled event preferences prevent new in-app notices for that event.
- Broadcast audience matches the selected department(s) or organization scope.
- Invalid, empty, or unauthorized broadcasts fail with a clear response and no partial delivery.



Application output: live in-app inbox, notification preferences, and Admin announcement controls.

03 Shift & work schedule management

Model working time as reusable shift definitions plus dated assignments, so schedule changes remain traceable and attendance metrics use the schedule that applied on the day.

Shift type	Configuration
General	Defined start/end times, expected duration, and grace period.
Night	Supports a checkout on the following calendar day.
Flexible	Uses the configured expected duration and applicable attendance policy.
Remote	Keeps scheduled work expectations while identifying the work mode.

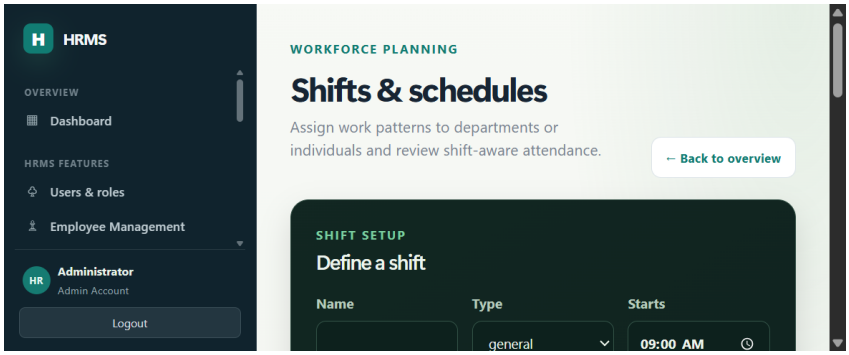
Assignment and calculation rules

An administrator assigns a shift to exactly one employee or one department for an effective date range. Assignment history is retained; a later assignment closes the preceding period rather than rewriting it. Where both department and employee records apply, the employee-specific assignment takes precedence. Attendance compares check-in with scheduled start plus grace, checkout with scheduled end, and actual duration with expected duration to derive late, early-checkout, and overtime minutes.

Reporting and overrides

Authorized HR/Admin users can filter attendance by date range, department, and shift. Admin corrections must identify the override, recalculate derived metrics against the selected shift, and record an audit reason. Ordinary employee check-in/out remains self-service and cannot modify schedule definitions.

Acceptance checks: reject invalid time ranges and overlapping/inconsistent assignments; test overnight shifts, grace boundaries, missing checkout, shift changes, department precedence, report filters, and audited corrections.



Application output: live shift setup, assignment history, and shift attendance report screen.

04 Advanced payroll & compensation

Keep compensation configuration, payroll calculation, and the employee-facing payslip clear and auditable. Every pay period must use the compensation terms effective for that period.

Layer	Required coverage
Base pay	Basic pay and configured fixed allowances such as HRA and special allowance.
Components	Earning and deduction components; fixed or variable amounts with effective dates.
Adjustments	Authorized bonus or other variable earning, and approved deductions, with an audit trail.
Attendance inputs	Working days, approved leave, loss-of-pay, and shift-derived overtime inputs.
Output	Period history and a payslip separating earnings, deductions, and net pay.

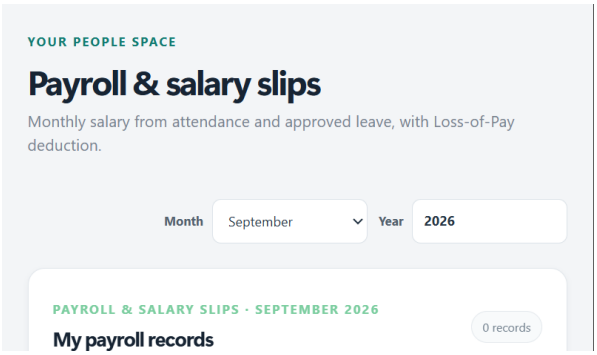
Calculation sequence

1. Select the compensation revision and component assignments effective in the payroll period.
2. Calculate period earnings and approved variable items.
3. Apply attendance-based proration and approved leave/loss-of-pay rules.
4. Apply configured deductions and adjustments.
5. Persist the payroll result and render the same stored breakdown on the payslip.

Controls and acceptance

Use decimal monetary values with explicit precision; reject negative values where not permitted and invalid effective-date ranges. A compensation revision closes the prior period and preserves history. Employees can access only their own payroll; team-wide configuration and payroll operations remain role-restricted. Re-running a period must not silently duplicate or overwrite an approved payroll result.

Decision required before monetary overtime is paid: approve the overtime rate, eligible roles, and rounding rule. Until approved, track overtime minutes and do not invent a pay multiplier.



Related application output: employee payroll and salary-slip workspace. This existing demo capture has no employee salary rows.

05 Security, data integrity & user experience

Apply consistent controls across API endpoints, database writes, and React workspaces. UI restrictions improve usability; backend authorization remains the security boundary.

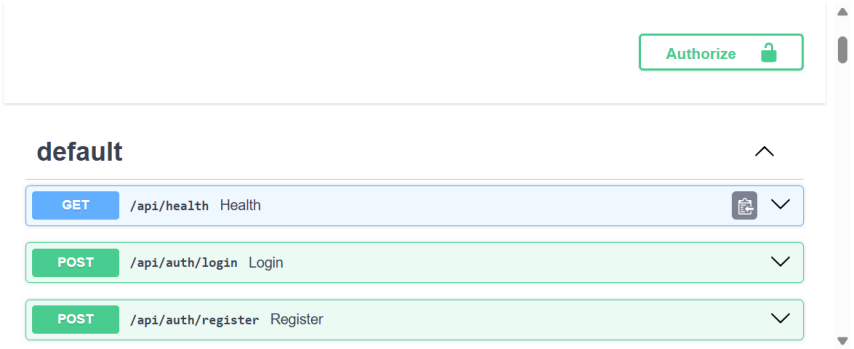
Area	Implementation requirement
Authorization	Admin/HR manage organization operations; employees read/update only their own permitted records.
Validation	Validate types, ranges, dates, department/employee ownership, duplicate records, and state transitions.
Auditability	Record actor, operation, target, timestamp, and reason for sensitive changes and overrides.
Data model	Keep user ownership explicit; index frequent user/date/status and employee/effective-date lookups.
Transactions	Commit each multi-record workflow atomically; roll back on constraint or delivery errors.
Responsive UI	Use accessible labels, visible validation/errors, compact mobile layouts, and keyboard-operable controls.

Operational considerations

Keep notification history user-scoped. Preserve shift and compensation revisions rather than overwriting operational history. Review database migrations against the existing SQLite development database and supported MySQL deployment schema. Use pagination or bounded date windows for growing inboxes and reports, and verify indexes against real query plans.

Release guardrails

No production credential or employee data belongs in generated documents or test fixtures. Migrations require a backup and rollback path. Email failures must not create a false success state for in-app delivery, or vice versa. API errors should be explicit and safe to display without leaking internal details.



API output: local Swagger UI documents the API operations and authorization controls.

06 Two-day implementation schedule

A two-day deadline is achievable as a focused integration and release-readiness window because the current codebase already contains Phase Two modules. It is not a reliable estimate for a greenfield build of all four workstreams.

Timebox	Priority work	Exit evidence
Day 1 Hours 0–1	Confirm scope, roles, current API/UI behavior, database state, and unresolved business rules.	Agreed acceptance checklist and risks.
Day 1 Hours 1–3	Close calendar/notification gaps; verify scope filtering, preference handling, and broadcast audience.	Focused API tests pass; no cross-user access.
Day 1 Hours 3–6	Verify shift assignment history, overnight calculations, report filters, and admin overrides.	Boundary and workflow tests pass.
Day 1 Hours 6–8	Verify salary revisions, component effective dates, attendance/leave inputs, payroll persistence, and payslip output.	Payroll reconciliation examples signed off.
Day 2 Hours 0–3	Integrate UI flows; review validation, responsive behavior, empty/loading/error states, and accessibility.	Role-based UI walkthrough complete.
Day 2 Hours 3–6	Run backend workflow tests, frontend build, migration checks, and API smoke tests.	Regression evidence and no critical failures.
Day 2 Hours 6–8	Fix release blockers, document setup/known decisions, confirm backup/rollback, and hand over.	Acceptance sign-off and release notes.

Critical-path decisions

Confirm optional-holiday attendance policy, overtime pay treatment, payroll period reprocessing rules, and who may broadcast. If these decisions are delayed, deliver tracking and reporting behavior first; do not infer financial policy.

07 Verification & completion checklist

Close the phase only when user-visible behavior, API permissions, calculations, and stored history agree. Evidence should come from repeatable tests and a short role-based walkthrough.

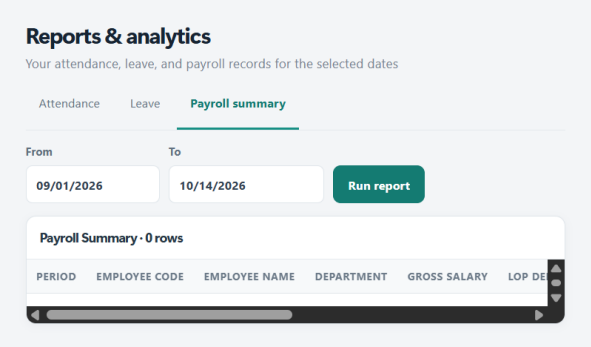
Workstream	Minimum acceptance evidence
Calendar	Company/regional/optional visibility, month/year navigation, and holiday-aware attendance summary.
Notifications	Event creation, user isolation, read/unread, preferences, and department/all-company broadcasts.
Shifts	Assignment history, employee/department scope, late/early/overtime boundaries, overnight shifts, and audited override.
Compensation	Effective-date revision, component totals, attendance/leave adjustment, payroll history, and payslip reconciliation.
Security/UI	Unauthorized API tests, cross-user isolation, responsive walkthrough, validation/error states, and production build.

Release handover

- Record the migration procedure, backup/rollback steps, and environment-specific settings.
- Publish the test/build results and note any remaining business-policy decisions.
- Confirm authorized Admin/HR and Employee walkthroughs without including live credentials.
- Confirm monitoring for API errors, notification delivery, payroll runs, and audit events.
- Obtain stakeholder sign-off against the acceptance evidence above.

DELIVERY OUTCOME
A secure, auditable workforce-management phase with clear holiday rules, user-specific engagement, shift-aware attendance, and explainable payroll breakdowns—delivered with verified behavior rather than scope claims alone.

Prepared 16 October 2026 from the supplied requirements and local repository review. This plan is not a production deployment certification.



Related application output: Reports & Analytics page from the local demo workspace.